

Actuarial Risk Analysis & Predictive Pricing Model

Transitioning from Exploratory Data Analysis to Predictive Modeling

Portfolio Size: 1,340 health insurance records

Methodology: SQL (Phase 1) | Multivariate Linear Regression (Phase 2)

Model Validation: 80/20 Train-Test Split | Residual Analysis

Final R-Squared: 78.09% (upgraded from 66.1% via interaction term)

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Classification: Academic / Portfolio Project

01 Executive Summary

This project explores a dataset of 1,340 health insurance records. My goal was to practice moving beyond basic historical averages to build a regression-based pricing model that can estimate expected claim costs for new applicants.

I completed this project in two phases. In Phase 1, I used SQL queries to extract descriptive statistics and identify initial risk patterns across geography, age, BMI, and lifestyle factors. In Phase 2, I used multivariate linear regression in Excel to quantify those patterns and produce a pricing formula.

Key findings from my analysis:

- Smoking status is by far the strongest predictor of claim cost in this portfolio.
- BMI is a meaningful secondary predictor, particularly when combined with smoking.
- Age and gender showed no statistical significance in this dataset ($P > 0.05$).

- Adding a "BMI x Smoker" interaction term improved the model's accuracy from 66.1% to 78.1% R-squared.
- Region (particularly the Northeast) contributed meaningfully to cost variation.

Project Limitation: Since I am working with a single sample dataset, this is more of a prototype than a production-ready model. The results may not generalize to different policyholder pools without revalidation.

02 Phase 1 - Exploratory Data Analysis (SQL)

Before jumping into regression, I used SQL to explore the raw dataset. I ran nine core queries to establish baselines and figure out which variables were worth investigating further. The most interesting findings are documented below.

2.1 Geographic Cost Distribution

I grouped claims by region to see if geography was a meaningful cost driver. Healthcare pricing often varies by location due to provider networks and local regulations.

```
SELECT region, COUNT(PatientID) AS total_patients, ROUND(AVG(claim), 2) AS
average_claim_cost FROM insurance_data GROUP BY region ORDER BY average_claim_cost
DESC;
```

REGION	TOTAL PATIENTS	AVERAGE CLAIM COST (\$)
Northeast	231	16,889.04
Southeast	443	13,058.52
Southwest	314	12,723.13
Northwest	352	11,583.30

The Northeast shows materially higher average claims (roughly \$5,300 above the Northwest). The sample size is smaller (n=231), which might explain part of the gap, but the difference was large enough that I decided to include regional dummy variables in my regression model. This instinct was validated later when the region returned a P-value of < 0.001.

2.2 Age Bracket Analysis

Going into this, I assumed age would be a massive primary cost driver. I grouped the dataset into five life-stage brackets to test that assumption.

```
SELECT CASE WHEN age BETWEEN 18 AND 24 THEN '18-24 (Young Adult)' WHEN age BETWEEN 25
AND 35 THEN '25-35 (Adult)' WHEN age BETWEEN 36 AND 45 THEN '36-45 (Middle Age)' WHEN
age BETWEEN 46 AND 55 THEN '46-55 (Senior Adult)' ELSE '56+ (Elderly)' END AS
age_bracket, COUNT(*) AS total_patients, ROUND(AVG(claim), 2) AS average_claim_cost
FROM insurance_data GROUP BY age_bracket ORDER BY average_claim_cost DESC;
```

AGE BRACKET	TOTAL PATIENTS	AVERAGE CLAIM COST (\$)
18-24 (Young Adult)	157	14,089.78
36-45 (Middle Age)	369	13,661.58
56+ (Elderly)	99	13,182.42
25-35 (Adult)	431	13,082.60
46-55 (Senior Adult)	284	12,541.54

Analyst Note: The Age Paradox

I found something really surprising here: the results don't follow the expected age gradient at all. The 18-24 year olds actually average higher claims than any other bracket. My theory is that there is a higher concentration of smokers in the younger cohort of this specific dataset, which inflates their average independently of their age. This was confirmed when my regression model returned a P-value of 0.939 for age, effectively ruling it out as a direct predictor. This taught me not to blindly trust standard assumptions!

2.3 High-Severity Claims & Intersectional Risk

To find the worst financial exposures, I filtered the dataset for claims above \$30,000 and cross-referenced them against BMI and smoking status.

```
SELECT age_bracket, bmi_category, smoker, COUNT(PatientID) AS high_cost_patients,
ROUND(AVG(claim), 2) AS average_massive_claim FROM insurance_summary_view WHERE claim >
30000 GROUP BY age_bracket, bmi_category, smoker ORDER BY average_massive_claim DESC
LIMIT 5;
```

AGE BRACKET	BMI CATEGORY	SMOKER	HIGH-COST PATIENTS	AVG. CLAIM (\$)
36-45 (Middle Age)	Obese	Yes	38	43,046.13
56+ (Elderly)	Obese	Yes	8	42,426.12
46-55 (Senior Adult)	Obese	Yes	29	41,306.83
25-35 (Adult)	Obese	Yes	50	41,173.29
18-24 (Young Adult)	Obese	Yes	19	40,633.71

Every entry in the top-five highest-severity segments shares the exact same two traits: Obese BMI and active smoker status. This strongly suggested to me that BMI and smoking do not operate independently in this portfolio, and that I needed to engineer a multiplicative interaction term for my regression.

2.4 BMI and Diabetic Status

A cross-tabulation of BMI category and diabetic status was run to assess whether diabetes compounds weight-related cost in a measurable way.

```
SELECT bmi_category, diabetic, COUNT(PatientID) AS total_people, ROUND(AVG(claim), 2)
AS average_cost FROM insurance_data GROUP BY bmi_category, diabetic ORDER BY
average_cost DESC;
```

BMI CATEGORY	DIABETIC	TOTAL PEOPLE	AVERAGE COST (\$)
Obese	No	377	15,589.98
Obese	Yes	331	15,465.94
Overweight	No	206	11,141.26
Overweight	Yes	183	10,841.16

Within each BMI category, diabetic and non-diabetic policyholders show near-identical average costs. The difference is less than \$200 in both the Obese and Overweight groups. This hinted that diabetic status may not add explanatory power once BMI is already in the model—a finding later confirmed by the regression ($P = 0.839$ for diabetic status).

03 Phase 2 - Predictive Regression Model

Next, I exported the data to Excel to build a multivariate linear regression model. I created dummy variables (1/0) for the categories, and set aside 20% of the data as a test set so I could honestly evaluate the model later.

3.1 Data Preparation

The 1,340 records were randomly split: 1,072 records (80%) were used to train the model, and 268 records (20%) were held back. I also engineered a "BMI x Smoker" interaction term by multiplying each policyholder's BMI value by their smoker binary flag.

3.2 Feature Selection (P-Value Pruning)

I ran the regression and looked at the P-values to see which variables actually mattered, dropping anything above a 0.05 significance threshold.

VARIABLE	COEFFICIENT (\$)	P-VALUE	DECISION
BMI x Smoker (Interaction)	+\$1,259	5.33e-65	Retained
Smoker Status (Base)	-\$17,828	< 0.001	Retained
Region Southeast	+\$3,644	< 0.001	Retained
Dependents (per child)	+\$738	< 0.001	Retained
Blood Pressure (per unit)	+\$159	< 0.001	Retained
BMI (per point)	+\$85.5	< 0.001	Retained
Diabetic Status	n/a	0.839	Dropped
Gender (Female)	n/a	0.232	Dropped
Age (per year)	n/a	0.939	Dropped

Interpreting the Smoker Coefficient

*The negative base coefficient for Smoker (-\$17,828) has to be read alongside the interaction term (+\$1,259 per BMI point). For a smoker with a BMI of 30, the combined smoking effect is: $-17,828 + (1,259 * 30) = +\$19,942$ in additional predicted cost.*

3.3 Pricing Formula

Based on the variables I kept, here is the pricing equation to estimate expected claims for new applicants:

Expected Claim =
-\$7,277 (Intercept)
+\$159.00 x Blood Pressure
+\$85.50 x BMI
+\$738.00 x Number of Dependents
-\$17,828 if Smoker (1=Yes, 0=No)
+\$1,259 x (BMI x Smoker)
+ Regional Adjustments (e.g., +\$3,644 for Southeast)

Calculation Example: For a Non-smoker, BMI 27, blood pressure 80, two dependents, living in the Southeast region:

$$\begin{aligned} & -7,277 + (159 \times 80) + (85.5 \times 27) + (738 \times 2) + 0 + 0 + 3,644 \\ & = -7,277 + 12,720 + 2,308.50 + 1,476 + 3,644 = \$12,871.50 \text{ estimated annual claim.} \end{aligned}$$

3.4 Model Validation

- **Training R-Squared (78.09%):** The model explains 78% of claim variance.
- **Residual Plot:** Showed a random scatter around zero, indicating the model isn't systematically over- or under-predicting.

04 Phase 3 - Business Recommendations

The following recommendations are based on the statistical findings above. Where relevant, regulatory constraints under the Affordable Care Act (ACA) have been factored in.

4.1 Smoker Premium Surcharge

Smoking is the single largest driver of claims in this portfolio, and it is one of the few risk factors that ACA community rating rules explicitly allow insurers to price on. A mandatory surcharge should be applied to all policies covering active smokers.

The surcharge amount should be derived from the pricing formula rather than a flat figure, since the interaction term shows that cost scales with BMI. For a smoker at BMI 25, the combined smoking penalty is approximately \$13,647 annually. At BMI 35, that rises to approximately \$26,237.

4.2 Secondary Underwriting Review for High-Risk Applications

The SQL query in section 2.3 identified that obese smokers consistently produce the highest-severity claims across all age groups. The underwriting platform should be configured to flag any application where *Smoker* = *Yes* and *BMI* > 30 for manual review before binding. This is a process control, not a pricing adjustment, so it does not raise the same regulatory concerns as differential premiums.

4.3 Marketing and Acquisition

Since age and gender did not show statistical significance in this portfolio, demographic targeting based on those factors is unlikely to improve risk selection. Budget should instead be directed toward lifestyle-based channels campaigns targeting non-smokers and physically active adults with low baseline blood pressure. Competitive discounting is viable for this segment given their predicted low claim costs.

4.4 ACA Compliance - Internal vs. External Use of Model Variables

Several variables that are statistically significant in the model—specifically BMI and blood pressure—are not permitted as direct pricing factors under ACA community rating rules. Adjusting consumer premiums based on these factors would expose the firm to regulatory risk.

The recommended approach is to use BMI and blood pressure internally for two purposes: reserve calculations (where the full model accuracy is needed) and targeted preventative-care outreach (e.g., mailing wellness programme information to high-risk policyholders). Consumer-facing

pricing should rely only on the ACA-permissible variables: smoking status, number of dependents, and geography.

05 Project Limitations & Future Learnings

As a portfolio project, this was a great learning experience, but I know there are limitations to how this model would work in the real world:

Unexplained variance (22%)	22% of the variance is likely driven by things not in my dataset—like genetics or accidental injuries. No regression model will perfectly predict human health.
Single dataset	Before applying this to a real business, the model would need to be retrained on a wider dataset.
Linear assumptions	Extreme medical outliers (like claims over \$60,000) might not be well-modeled by a linear approach. Learning to use a GLM framework is a personal next step for me.
Blood pressure tracking	The dataset only provides point-in-time readings, so the model can't distinguish between chronic hypertension and someone who was just nervous at the doctor.